ABHI AREKERE

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EDUCATION

The University of Texas at Austin, Cockrell School of Engineering, Austin, TX

Bachelor of Science, Biomedical Engineering — Cellular and Molecular Track *Graduated May 2026*

- Relevant Coursework: Biomedical Instrumentation, Signals & Systems Analysis, Biochemistry, Cell Engineering, Computing in Python

TECHNICAL SKILLS

Equipment & Instrumentation: laboratory equipment troubleshooting, preventive maintenance, calibration, verification testing, equipment performance checks; incubators, centrifuges, *Drosophila* Activity Monitors, fluorescence/confocal microscopes

Laboratory: PCR, gel electrophoresis, mammalian cell culture, western blotting, immunostaining, BSL-1/BSL-2

Software & Data: Python (pandas, NumPy, plotly), R, dbt, Git, Linux/Unix, Excel, Word, MATLAB, SQL, PostgreSQL, Redshift

Genomics & Sequencing: nf-core/Nextflow, BWA-MEM, GATK, FASTQ/BAM/VCF, snRNA-seq analysis (Seurat, Signac, Monocle3)

WORK EXPERIENCE

Research Assistant, Neurogenetics Laboratory *May - Aug 2025; May - Aug 2023*

Keene Lab, Texas A&M University, College Station, TX

- Maintained 24 *Drosophila* Activity Monitors and 5 incubators within specification through scheduled checks and logged maintenance; diagnosed and resolved channel failures, cabling issues, and data acquisition errors
- Phenotyped six genotype-by-treatment groups of humanized APOE3/APOE4 *Drosophila*; work contributed to a 2026 bioRxiv preprint
- Cleaned, normalized, and analyzed snRNA-seq data in R, from QC filtering through clustering and trajectory inference

Research Intern, Tumor Microenvironment Laboratory *May - Aug 2024*

Sarkar Lab, Texas A&M University, College Station, TX

- Quantified nanoparticle therapeutic response in treated and untreated 4T1 murine triple-negative breast cancer and 3T3 fibroblast cultures using western blotting, scratch wound assays, and fluorescence microscopy, with quantitative image analysis

Research Assistant (Remote), Genome Evolution Laboratory *Jul 2020 - May 2023*

Blackmon Lab, Texas A&M University, College Station, TX

- Built R pipelines that integrated karyotype data with 1.46 million GBIF species occurrence records spanning 1,023 Carnivora species, yielding a phylogeny-matched analysis set of 110 species; work contributed to a peer-reviewed publication in the *Journal of Heredity*

PROJECTS

Clinical Genomics Workflow Monitoring | Industry-Sponsored Senior Design Project *Aug 2025 - May 2026*

The University of Texas at Austin, Austin, TX

- Built a simulator of Natera's Signatera sample processing workflow, generating sample-level events with QC failures and rework loops
- Defined workflow KPIs (dwell time, queue delay, throughput, rework rate) to flag bottlenecks and potential turnaround time delays
- Developed a data pipeline loading workflow events into PostgreSQL and Amazon Redshift, transformed through layered dbt models
- Deployed a linear regression model in Redshift ML to predict case turnaround time, achieving a test MAE of 6.3 hours and MAPE of 14.6% on simulated workflow data
- Built a React dashboard with a FastAPI backend displaying sample status, SLA risk, turnaround predictions, and intervention flags to operators; presented platform to Natera leadership

SolarSense: BLE-Connected UV Dosimeter | BME Design Project *Jan 2025 - May 2025*

The University of Texas at Austin, Austin, TX

- Built and bench tested a wearable UV dosimeter, integrating electronics, a SOLIDWORKS enclosure, ESP32 firmware, and an iOS app
- Verified UV readings within 10% of reference, 15m BLE range, and 8h of runtime; passed 7 of 8 verification tests against requirements
- Documented design requirements, acceptance criteria, verification protocols, test results, and risk analysis in a design history file to support traceable engineering verification

Germline Variant Calling Benchmark | Independent Project *Aug 2026 - Present*

- Running a containerized Nextflow pipeline (nf-core/sarek, Docker) that turns raw sequencing files into benchmarked variant calls
- Validating pipeline output against the GIAB v4.2.1 truth set with rtg vcfeval and reviewing FastQC/MultiQC data quality reports

PUBLICATIONS

- Jonika, M.M., Wilhoit, K.T., Chin, M., **Arekere, A.**, and Blackmon, H. Drift drives the evolution of chromosome number II: The impact of range size on genome evolution in Carnivora. *Journal of Heredity*, 2024, 115(5): 524-531. https://doi.org/10.1093/jhered/esae025
- Biglari, S., …, **Arekere, A.,** …, Keene, A.C. Progressive behavioral and cognitive decline in *Drosophila* harboring AD-associated APOE4 variants. bioRxiv, 2026 (preprint). https://doi.org/10.64898/2026.09.08.750261